Mantra Outtandy

Final-year engineering student in Data & AI | Reinforcement learning, neural combinatorial optimization

Paris, France | linkedin.com/in/mantra-o-b0a99b24a | github.com/Mantra20

Education

ESILV Paris, Engineering degree (master's level), major in Data and Artificial Intelligence — expected Oct. 2027
Generative models (VAEs, GANs, diffusion), transformers and LLMs, deep learning, optimization, numerical methods.

Université Sorbonne Paris Nord, DUT in Data Science — 2022 – 2024

Research Experience

NAVER Labs Europe, Research collaboration, neural combinatorial optimization — Sept. 2026 – present, remote
Project lead: I set the research direction and coordinate a 3-person team.

- Refactored the PyTorch codebase from fixed start and end nodes to variable cost pairs.
- Implementing a MuZero-style latent transition model, trained by supervised learning, that predicts the next state embedding and removes one instance re-encoding per MDP step.
- Benchmarking sparse against quadratic attention on TSP and Job Shop Scheduling, measuring solution quality versus training time and generalization to unseen instance sizes.
- Co-applicant on a GENCI national GPU allocation; experiments run with SLURM on A100 nodes.

Independent research, Conserved quantities in self-supervised video models — 2026, github.com/Mantra20/eb_jepa
Built on the facebookresearch/eb_jepa codebase.

- Tested whether a video JEPA encodes kinetic energy and momentum, with a probing protocol including a permutation control and a triviality test.
- A quadratic probe reaches $R^2 = 0.75$ while a linear probe fails; causal ablation via modified training code (300 epochs per run on A100).

Professional Experience

Thales, Land and Air Systems, RL Research Engineer (apprenticeship) — Sept. 2025 – Sept. 2027, Paris

- Wrote the state of the art on RL for decision making in resource allocation.
- Built a custom Gymnasium simulation environment from scratch, including object propagation, physical trajectories and constraints.
- Implemented RL algorithms and a benchmark suite scoring them against heuristic and deterministic baselines.
- Trained supervised predictive models and generated training scenarios from operational data with RL (under NDA).

French Ministry of the Armed Forces, Data Scientist (apprenticeship) — Sept. 2024 – Sept. 2025, Paris

- Built predictive maintenance models on 8,400 vehicles, flagging failures 6 to 8 weeks ahead and reducing unplanned downtime by 23%.
- Designed a cost model that cut budget overestimation from 18% to 4%; automated the data and training workflows.
- Ran 4 workshop series for 45 analysts across 3 regional commands, in French and English.

Selected Projects

- **SimForge** (github.com/leogue/simforge): CAD-to-simulation pipeline exporting URDF, MJCF and USD for MuJoCo, Isaac Sim and Genesis. Wrote the validation layer with Pinocchio (forward kinematics, Jacobians, RNEA torques).
- **Flocking agent** (github.com/Mantra20/boid_rl_agent): DQN from scratch in PyTorch (network, replay buffer, target network, training loop), evaluated against Q-Learning and SARSA.
- **Proteus**, co-founder and Vice-President: first French student defense-technology federation, 3 schools, 4 technical projects, partnerships with Thales and the French defence procurement agency.

Skills

Machine learning	Transformers, sparse attention, MuZero, latent world models, deep RL
Tools	Python, PyTorch, C++, Gymnasium, MuJoCo, Git, SLURM (A100 clusters)
Languages	French (native), English (C1), Tamil